Yash Siddhapura

Electronics & Communication Engineering

L.D. College Of Engineering

siddhapurayash09@gmail.com | +91 7284985820 | Yash Siddhapura | Siddhapura-Yash

EDUCATION

- **B.E. in Electronics & Communication** **2023-26**
  L.D. College of Engineering, Ahmedabad
  CGPA : 8.28/10

- **Diploma in Electronics & Communication** **2020-23**
  A.V. Parekh Technical Institute, Rajkot
  CGPA : 9.28/10

WORK EXPERIENCE & INTERNSHIPS

- **FPGA Engineer | Vicharak** *(Jan'26 - present)*
  - Designed the streaming architecture of an FPGA-based stereo vision accelerator by studying existing stereo-matching approaches and research papers, incorporating BRAM-efficient, high-throughput processing for real-time depth estimation.
  - Implemented key RTL modules in Verilog, including cost computation, cost aggregation, disparity estimation, and depth generation, validated at 140 MHz on FPGA hardware.
  - Replaced vendor FIFO IP with custom synchronous and asynchronous FIFOs, achieving 151 MHz operation across multiple clock domains.
  - Implemented and verified UART, I2C, and UART-to-DDR data transfer pipelines in Verilog on FPGA.
- **Intern | Power Electronics** *(July'23 - Sep'23)*
  - Worked on transformer winding and testing, gaining hands-on experience in power electronics.
  - Developed skills in electrical component analysis and circuit testing.

PROJECTS

- **CNN Accelerator for Real-Time Handwritten Digit Recognition**
  - Designed and implemented an INT8 CNN accelerator in Verilog for FPGA-based handwritten digit recognition (MNIST).
  - Developed a fully hardware inference pipeline with convolution, ReLU, max-pooling, and fully connected layers, verified on the Ti375 Efinix FPGA.
  - Achieved 100 MHz operating frequency with 16 parallel INT8 MAC units, delivering 3.2 GOPS (0.0032 TOPS) peak compute throughput.

- **32-bit ARM-inspired RISC Processor in Verilog**
  - Designed and implemented a 32-bit ARM-inspired RISC CPU in Verilog using modular RTL design.
  - Developed core components including ALU, register file, datapath, control unit, instruction decoder, and memory interface.
  - Supported arithmetic, logical, and memory instructions (ADD, SUB, MOV, CMP, B, LDR, STR) with verified fetch-decode-execute operation on FPGA.

- **Master-Slave Robotic Arm**
  - Built a 4-DOF master-slave robotic arm with motion recording and playback.
  - Implemented real-time motion replication, enabling the slave arm to accurately mimic the movements of the master arm.
  - Developed record-and-playback functionality to store and repeatedly execute predefined arm trajectories for pick-and-place operations.

TECHNICAL SKILLS

- **Programming Languages** : Verilog, SystemVerilog, VHDL, C, C++, Python, Linux
- **Simulation & EDA Tools**: Modelsim, Quartus, Vivado, EDA Playground, iverilog, Effinity, Proteus

ACHIEVEMENTS & AWARDS

- Awarded a ₹30,000 SSIP Grant for the Master-Slave Robotic Arm project.
- Achieved Rank 50 (Gujarat, all branches) in B.Tech Lateral Entry admissions, 2023.
- Strengthened RTL design and logic fundamentals by solving 141+ verilog challenges on HDLBits.